

Pareto Optimality

Let A be a set (of alternatives) and $\{\succeq_i\}_{i=1}^I$ a family of preference (i.e., complete and transitive binary) relations on A . An alternative $a'' \in A$ *Pareto dominates* $a' \in A$ if $a'' \succeq_i a'$ for $i = 1, \dots, I$ and $a'' \succ_j a'$ for some $j \in \{1, \dots, I\}$. The alternative $a^* \in A$ is *Pareto Optimal* if there is no $a \in A$ which Pareto dominates it.

Suppose that, for each $i = 1, \dots, I$, $\succeq_i$ is represented by a function u^i on A . Let $W : \mathbb{R}^I \rightarrow \mathbb{R}$ be *strictly increasing*.¹

Lemma. *Any solution to*

$$\max_{a \in A} W(u^1(a), \dots, u^I(a)) \quad (1)$$

is Pareto Optimal.

Proof: I will show that a point which is not Pareto Optimal does not solve (1). Suppose that $\hat{a} \in A$ is not Pareto Optimal. Then there is an $a' \in A$ with

$$u^i(a') \geq u^i(\hat{a}) \quad (2)$$

for $i = 1, \dots, I$ and

$$u^j(a') > u^j(\hat{a}) \quad (3)$$

for some $j \in \{1, \dots, I\}$. Since $W(\cdot)$ is strictly increasing $W(u^1(a'), \dots, u^I(a')) > W(u^1(\hat{a}), \dots, u^I(\hat{a}))$. So $\hat{a}$ does not solve (1). $\square$

A special case is the function $W(u) = \sum_i \lambda_i u^i$, where $\lambda_i > 0$ for $i = 1, \dots, I$. We will use this form for $\lambda_i = 1$ for $i = 1, \dots, I$.

¹ $W(u') > W(u)$ for any u', u in $\mathbb{R}^I$ with $u' > u$.